Kerr-ECT Coupling:

How Fixed-Chiral Quantum Torsion Stabilizes Mass from Micro-Particles to Black Holes

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Abstract

Zou (2025), based on the MITC (Multiplicative-Entropy-Driven Invariant-Topology Chiral-Space) model, established a unified framework spanning from the quantum scale to the cosmic scale. The core contributions of that work include: (1) defining mass as the elastic strain energy stored in the spatial SEQ (Space Elementary Quanta) network under compression by SU(3) color interactions, frozen into a stable bound state via a Higgs-like chiral ratchet-lock coupling; (2) elucidating the physical mechanism by which mass induces a gravitational field by stretching the surrounding compressed space, providing a geometric intuition for the mass-gravity duality; (3) interpreting dark matter as a quasi-compressed state locked by the spinor field of the spatial network—its locking strength is lower than that of ordinary mass states, yet sufficient to produce observable gravitational effects, thereby naturally explaining galaxy rotation curve anomalies without the fine-tuning problems associated with introducing unknown particles.

Building upon the above work, this paper, in the form of a review, further focuses on the necessary extension of the "spinor-locked mass" principle within the MITC model to macroscopic scales. This paper argues that: the rotating spacetime geometry described by the Kerr metric is, in essence, the macroscopic continuum limit manifestation of the SEQ resonance axis direction field (i.e., the spinor field of the spatial network) in the MITC model; and the quantum torsion field defined in the Einstein-Cartan theory (ECT) is, within this framework, redefined as the continuum field-theoretic expression of the fixed-chiral ground-state spin of the SEQ. When these two are coupled through the topological locking potential in the MITC model, they constitute the universal condition for mass stability. This coupling mechanism endows Kerr geometry with an active mass-locking function, rather than merely a passive kinematic description.

The core deduction derived from this is: all stable macroscopic mass bodies—from hadrons to stellar-mass black holes and up to supermassive galactic black holes—must rely on a non-zero spatial network spinor field to maintain the persistence of their mass states. This implies that, within the MITC model framework, a Schwarzschild black hole with pure radial compression and no spinor locking cannot exist stably in physical reality; all real black holes must be Kerr-type (rotating) black holes. This deduction is consistent with all black hole observational data to date (all showing non-zero spin) and provides a clear falsifiable criterion for future gravitational wave and black hole shadow observations: if a compact massive astrophysical object with zero spin angular momentum is discovered, this extended deduction of the MITC model will be directly falsified. This paper aims to systematically review the existing discourse on spinor-mass binding in the MITC model and focus on elaborating its necessary deductions and testable predictions in the field of black hole physics.

1. Introduction

1.1 The Problem of Mass Persistence

In Einstein's general relativity, mass curves spacetime through the Einstein field equations. These equations precisely describe how mass deforms spacetime geometry—whether in the radially symmetric Schwarzschild solution or the rotating Kerr solution. However, the Einstein equations only provide a static description of "given a matter distribution, what the geometry is," yet they never explain **why these geometric deformations can persist**.

In general relativity, spacetime curvature is a purely geometric property that does not carry a concept of strain energy density or elastic restoring potential. The curvature tensor purely describes the degree of geometric bending **and does not include a dynamical term such as "the more intense the bending, the stronger the restorative force."** Therefore, GR can only **describe** the geometric form of Schwarzschild or Kerr black holes, but cannot **explain** why these curved configurations can persist without relaxing back to flat spacetime.

In the Standard Model of particle physics, the Higgs mechanism bestows mass upon elementary particles via Yukawa couplings, and QCD chiral condensation imparts mass to hadrons. However, these mechanisms are applicable only at the quantum scale. For macroscopic mass bodies—especially black holes with masses exceeding that of the Sun—the physical mechanism responsible for their mass stability remains an open question.

1.2 Limitations of Existing Frameworks: Describing Deformation, Not Explaining Locking

General relativity accurately portrays the spacetime geometric deformations induced by mass, but **it has never explained why these geometric deformations can persist**. In an elastic space picture, any compression is accompanied by a tendency to rebound; without an additional locking mechanism, spacetime curvature should rapidly relax back to a flat state. The field equations of GR constitute a **static taxonomy**, not a **dynamic theory of stability**. It is precisely this lacuna that provides the logical space for introducing Kerr-ECT coupling within the MITC model.

1.3 Foundational Contributions of the MITC Model

Zou (2025) proposed the MITC model, providing a foundational framework for the above problem:

- Space is composed of a topologically fixed network of SEQs, with energy transferred discretely between adjacent nodes in units of Planck's constant;
- Mass is the elastic strain energy stored in the SEQ network under SU(3) compression;
- A Higgs-like mechanism acts as a "chiral ratchet lock," fixing the compressed state so that it does not rebound;

- Dark matter is essentially a **quasi-compressed state** locked by the spinor field—with locking strength lower than ordinary mass, yet sufficient to produce observable gravitational effects, thereby explaining galaxy rotation curve anomalies;
- Mass generates a gravitational stretching of the spatial network, which is the physical origin of the gravitational field.

Building on this, this paper proposes a general extension: **Kerr geometry + ECT quantum torsion + fixed-chiral coupling locking = a universal mass stabilization mechanism extending from the Higgs mechanism for micro-particle mass stability to black holes.**

1.4 Central Thesis of This Paper

The central thesis of this paper can be stated as:

> **Mass Stability = Spatial Compression × Spinor Locking**
>
> Where "spatial compression" is achieved by SU(3)/QCD at the microscopic scale or by gravitational collapse at the macroscopic scale; "spinor locking" is accomplished by the fixed-chiral coupling between the macroscopic spinor field described by the Kerr metric and the quantum torsion defined by ECT.

The necessary deduction from this is: **all stable black holes must carry non-zero spin, i.e., must be Kerr black holes.**

2. Review of Core Elements of the MITC Model

2.1 SEQ Network and Spatial Elasticity

In the MITC model, space is composed of a topologically fixed discrete network of SEQs. Adjacent SEQs are connected by sub-Planckian elastic bonds, constituting a mechanical structure akin to a spring network. Spatial deformation (compression or stretching) corresponds to the storage of elastic potential energy. Energy, in integer multiples of Planck's constant, is irreversibly transferred along energy gradients from high-energy nodes to low-energy nodes.

2.2 Mass as a Compressed State and Higgs Locking

When a local region of the SEQ network is compressed under SU(3) color interactions, elastic potential energy is stored. However, mere compression is unstable—like a compressed spring, it will rebound once the external force is removed. The MITC model proposes that a Higgs-like mechanism provides a **"chiral ratchet lock"**:

Chiral: The coupling depends on the orientation of the SEQ's fixed chiral spin;

Ratchet: Compression energy can only be stored in one direction; the reverse rebound is geometrically forbidden;

Locking: The compressed state is frozen into a persistent mass.

This locking mechanism relies on the spatial network spinor field—the local torsion of the SEQ resonance axis direction field.

2.3 Dark Matter as a Quasi-Locked State

Zou (2025) points out that dark matter is not unknown particles, but rather **quasi-compressed spatial states with locking strength lower than that of ordinary mass states**. Its formation conditions are:

- The spatial compression intensity is insufficient to reach the threshold for mass-state locking;
- However, there is still enough network spinor coupling to keep the compressed state in a metastable locked state;
- This quasi-locked state can still stretch the surrounding space, producing observable gravitational effects.
- This explanation naturally accounts for why dark matter only participates in gravitational interactions and has no electromagnetic signals, while avoiding the fine-tuning of parameters associated with introducing new particles.

2.4 Mechanism of Mass-Induced Gravitational Field

In the MITC model, the physical picture of mass inducing a gravitational field is:

- Local space is compressed and locked → forming a mass source;
- The mass source produces a **stretching** effect on the surrounding SEQ network;
- This stretching propagates outward as spherical waves, manifesting as a gravitational field;
- The degree of stretching attenuates according to the inverse square law (due to spherical flux conservation).
- This naturally aligns with Newton's law of gravitation and the weak-field limit of general relativity, while providing a physical intuition that the gravitational field originates from the elastic strain of space.

3. Kerr Metric and the Geometric Expression of Spatial Spinors

3.1 The Kerr Metric as a Macroscopic Manifestation of the Spinor Field

The Kerr metric describes the spacetime geometry of an axisymmetric black hole with angular momentum. A key geometric feature is **frame-dragging**—the rotating mass drags the surrounding spacetime, forming a closed spinor field structure.

In the MITC model, this macroscopic spinor field corresponds to the macroscopically ordered arrangement of the **resonance axis direction field** of a large number of SEQs, i.e., the macroscopic continuum limit manifestation of the SEQ resonance axis direction field. The spinorial structure of Kerr spacetime is simply the continuum limit manifestation of the SEQ network spinor field.

3.2 Limitations of the Mainstream View on the Spinor-Mass Relationship

In the mainstream understanding, the spinor field in the Kerr metric (i.e., the frame-dragging effect) is a **consequence** of the mass's rotation, not the **cause** of the mass's stability. This understanding reduces the relationship between mass and spinor to a **coexistence phenomenon**, lacking a mechanistic causal chain.

In other words: The mainstream view would say "mass rotates, therefore it produces Kerr geometry"; this paper argues that "spinor locks mass, therefore a stable mass must rotate."

3.3 Instability of Pure Radial Compression

In the absence of a spinor field, i.e., under pure radial compression (Schwarzschild type), the SEQ network is in a state of isotropic compression. Within the elastic framework of the MITC model, this state cannot trigger the locking potential due to the lack of chiral boundary conditions—a compressed state without spinor locking will inevitably become unstable under elastic rebound and relax. Physically, this means that compressed space without spinor locking will undergo elastic rebound oscillations, failing to form a persistent bound state.

4. Einstein-Cartan Torsion: Field-Theoretic Expression of SEQ Fixed-Chiral Spin

4.1 From SEQ Spin to Continuous Torsion Field

In the MITC model, each SEQ carries a **fixed-chiral ground-state spin**. This spin manifests in the discrete network as the **intrinsic chiral orientation** of the SEQ resonance axis vector, invariant with respect to energy transfer or spatial deformation.

When taking the continuum limit, the discrete SEQ spin orientation field maps to a **local director field** on the spacetime manifold. The spatial gradient of this local director field yields a torsion-type structure, which is defined in the MITC model as the **continuum field expression of**

the SEQ fixed-chiral spin. We refer to it as the **ECT torsion field**—however, this nomenclature borrows only its mathematical structure; its physical content is entirely defined by the collective behavior of the SEQ ground-state chiral spins.

4.2 Functional Role of ECT Torsion in the MITC Model

Function	Description
Providing a chiral background	Provides a directional criterion for locking the compressed state of the SEQ network—only spinor configurations matching the fixed chirality can be locked
Defining local torsion density	Describes the distribution of SEQ spin orientation deviations in space, i.e., the degree of "twist" of the spatial network
Constituting a variable for the locking potential	Together with the macroscopic spinor field of the Kerr metric, forms the two independent variables of the locking potential

In this framework, the ECT torsion field assumes the following functions:

4.3 Distinction from Traditional ECT

The term "ECT torsion" as used in this paper borrows only its mathematical form. Its physical meaning is redefined within the MITC model:

Comparison	Traditional ECT	ECT Torsion in the MITC Model
Physical Origin	Intrinsic spin density of fermions	Fixed-chiral ground-state spin of SEQs
Geometric Role	Antisymmetric part of the affine connection (torsion)	Gradient of the SEQ resonance axis direction field
Function	Describes spin-spacetime coupling	Provides the chiral boundary conditions necessary for mass locking

4.4 Summary

In the MITC model, the ECT torsion field is reinterpreted as the **collective field-theoretic manifestation of the fixed-chiral SEQ spin in continuous spacetime**. It does not carry the physical content of spin-torsion coupling in traditional ECT, but instead provides **microscopic chiral anchors** for the macroscopic Kerr spinor field—and it is precisely this anchor that enables the compressed spatial state to be frozen into persistent mass via the locking potential.

5. Kerr-ECT Coupling: A Unified Mechanism for Mass Stability

The previous chapter clearly defined: the ECT torsion in this framework refers to the continuum field expression of the fixed-chiral ground-state spin of the SEQ. Building upon this, this chapter couples it with the macroscopic spinor field described by the Kerr metric through the locking potential.

5.1 Formal Expression of the Coupling

We propose **Kerr-ECT coupling**, i.e., the coupling between the macroscopic spinor field in the Kerr metric (frame-dragging) and the quantum torsion field in ECT (SEQ fixed-chiral spin) through a locking potential. This locking potential is a function of the Kerr spinor field and the ECT torsion field, behaving in the low-energy limit as a monotonically increasing function of the coupling strength between them. The potential takes its minimum value when both the macroscopic spinor and quantum torsion are non-zero, mathematically expressing that "a non-zero spinor is a necessary condition for mass locking."

5.2 Energy Landscape of the Locking Mechanism

The energy landscape of this coupling is as follows:

State	Spatial Compression	Spinor Field	Locking Status	Result
Flat Space	None	Zero	Not applicable	No mass
Schwarzschild Type	Yes	Zero	**Cannot lock**	Oscillatory rebound, no stable mass

State	Spatial Compression	Spinor Field	Locking Status	Result
Kerr Type	Yes	Non-zero	**Locked**	Stable mass body (from hadrons to black holes)
Dark Matter State	Weak	Weakly non-zero	**Quasi-locked**	Metastable, produces only gravity
Baryon (Hadron)	Strong (QCD compression)	Non-zero (quark-level spinor)	**Locked**	Stable baryon mass
Lepton (e.g., electron)	Radial compression negligible, approaching zero	Non-zero (pure spinor twist)	**Locked**	Small mass (originating from pure spinor twist)

This table intuitively illustrates two points: (1) **The presence or absence of a spinor field is the determining factor for the existence of a stable mass state**; (2) The origin of lepton mass differs from that of baryon mass—baryon mass relies on strong compression plus spinor locking, while lepton mass primarily originates from the small mass provided by pure spinor twist, with the radial compression contribution approaching zero.

5.3 Unity from Microscopic to Macroscopic Scales

The manifestation of this mechanism across different scales is consistent:

Scale	Compression Source	Spinor Locking Source	Mass Body
Quantum Scale	SU(3)/QCD	Quark-level structural spinor	Hadrons (proton, neutron, etc.)
Mesoscopic Scale	Electroweak interaction	Higgs coupling spinor	Leptons

Scale	Compression Source	Spinor Locking Source	Mass Body
Macroscopic Scale	Gravitational collapse	Kerr spinor (frame-dragging)	Black holes (stellar-mass and above)
Cosmological Scale	Large-scale structure compression	Dark matter quasi-locking spinor	Galactic halos (dark matter)

Each row of this table follows the same underlying logic: **Spatial Compression + Spinor Locking = Stable Mass**. The lepton case corresponds to the special situation where "radial compression is negligible, and spinor locking provides the primary mass," but it still does not violate the unified logic above.

6. Core Deduction: All Stable Black Holes Must Be Kerr Black Holes

6.1 Logical Chain from Theory to Deduction

Based on the preceding analysis, the logic of the deduction is as follows:

- Mass is the elastic strain energy of compressed space and must be stabilized through spinor locking;
- The Kerr metric is the geometric expression of the macroscopic spinor field;
- Therefore, any persistent macroscopic mass body must possess a Kerr-type spinor structure;
- Black holes are the most extreme macroscopic mass bodies, where the dependence of mass stability on spinor locking is the strongest;
- Consequently, all stable black holes must carry non-zero angular momentum—i.e., must be Kerr black holes.

6.2 Relationship to the Existing "No-Hair Theorem"

This deduction is **not a repetition** of the no-hair theorem, but rather a **deepening and complement** of it:

Comparison Dimension	No-Hair Theorem	This Deduction (Kerr-ECT Coupling)
Content	Stationary black holes are	Stable black holes must have

Comparison Dimension	No-Hair Theorem	This Deduction (Kerr-ECT Coupling)
	described by mass, angular momentum, and charge	non-zero angular momentum
Nature	Kinematic classification	Dynamic-thermodynamic necessity
Explanatory Level	Descriptive	Causal
Permissibility of Zero Angular Momentum	Theoretically allowed (Schwarzschild is a special solution)	Physically forbidden (cannot lock mass)

6.3 Consistency with Observational Facts

All confirmed black hole candidates to date have been confirmed or inferred to possess non-zero spin. No observation has yet found a black hole with definitively zero spin angular momentum.

This model elevates this "observational fact" to a "theoretical necessity": **even if the initial collapse is perfectly symmetric, the final stable black hole will inevitably acquire spinor through some mechanism due to the need for mass locking**—possibly through angular momentum transfer from an accretion disk, or through spinor extraction and re-locking via the Penrose process.

6.4 Falsifiability

This model simultaneously provides clear falsification conditions:

> **If a future observation discovers a compact object with a mass exceeding three solar masses, whose spin angular momentum is zero and this state is confirmed to be intrinsic rather than a result of accretion dilution, then the Kerr-ECT coupling locking mechanism of this model will be falsified.**

This falsifiability ensures the **scientific** nature of this paper, rather than mere metaphysical speculation.

7. Unified Explanation of Dark Matter

The explanation of dark matter within this framework can be directly incorporated:

Dark matter is a state where **spatial compression is insufficient** to reach the threshold for mass locking;

However, there is still **enough spinor coupling** to keep it in a **quasi-locked** state;

This quasi-locked state has a **weak elastic restoring force**, yet its stretching effect (gravity) on the surrounding space remains considerable;

Its mass density distribution is determined by the spinor field configuration in the quasi-locked region, naturally forming a halo-like distribution on galactic scales.

This explanation avoids introducing unknown particles and also circumvents the fine-tuning parameter problems associated with fields like axions.

8. Conclusion and Outlook

8.1 Summary

The main contributions of this paper can be summarized in three points:

Mechanistic Unification: Proposes Kerr-ECT coupling as a general mechanism for mass stability, coupling the macroscopic spinor geometry of the Kerr metric with the quantum torsion field of ECT through a fixed-chiral locking potential, achieving full-scale coverage from micro-particles to black holes.

Causal Upgrade: Elevates the relationship between mass and spinor from a "coexistence phenomenon" to a "causal binding"—mass stability **depends on** spinor locking, rather than merely "accompanying" the spinor.

Explicit Deduction: Derives the necessary conclusion that all stable black holes must be Kerr black holes, and provides a falsifiable observational criterion for this conclusion—the discovery of a compact massive object with zero spin angular momentum would directly falsify this model.

8.2 Relation to Zou (2025)

This paper is a natural extension of Zou's (2025) MITC model into the domain of macroscopic gravity:

MITC provides the underlying SEQ network and energy gradient dynamics;

MITC explains the cooperative generation of mass by QCD+Higgs and its induction of gravitational fields;

MITC interprets dark matter as a quasi-locked spatial compression state;

This paper further argues that Kerr geometry + ECT torsion coupling is precisely the explicit expression of the MITC locking mechanism at the macroscopic black hole scale.

8.3 Future Research Directions

- Conduct numerical simulations of the rebound process of a pure radial compression (zero angular momentum) mass state to verify the timescale of its instability;
- Investigate how the accretion process affects the spinor locking state of black holes to explain the diversity of observed spin parameters;

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